Summary of paper 02: carryover representation and inference for biomarker-treatment interaction tests with a predictive biomarker in aggregated N-of-1 trials under the correlation encoding of the interaction

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November 07, 2026

Problem and approach

Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Drug carryover attenuates the biomarker-by-treatment interaction. The analyst must decide how to represent residual exposure, how to obtain a correctly sized test, and whether the half-life must be known. We ran an exploratory simulation study under the recommended DGP, in which the interaction is a correlation between the biomarker and the drug response and the biomarker is predictive only. An analysis ladder changes one thing per rung: the starting analysis E0 (rung 0), a binary indicator with residual AR(1) correlation G1 (rung 1), CR2 inference G6 (rung 2), an exposure-weighted predictor at the true half-life G8 (rung 3, an oracle benchmark), and the half-life chosen by AIC on the model without the interaction G10 (rung 4). The side analysis G9 chooses the half-life with the interaction in the model. A biomarker that is also prognostic is the subject of a companion paper (1).

Simulation design

Table 1 lists the six scenario families.

Table 2 defines the ten G-family specifications.

Results

The ladder at a glance

Table 3 defines the ladder.

Figure 1 gives size by family.

Table 1: The six scenario families of the ladder comparisons. Every biomarker is predictive only. Within a cell every analysis is fitted to the same simulated trials; G10 was fitted to the stored trials of families 1 to 5 afterward. Family S has independent seeds for every cell. the full report describes the cells.

Family	Scenarios	Replicates
1	Base 30-cell grid: three designs, $N \in \{35, 70\}$, $c_{bm} \in \{0, 0.30, 0.45\}$, $t_{1/2} \in \{0, 0.5, 1\}$ weeks	500
2	Misspecified half-life (0.2, 2 or 4 weeks) and decay shape (Weibull $k \in \{0.5, 2\}$), analyzed as exponential	500
3	Serial correlation $\rho \in \{0.5, 0.9\}$	500
4	Patient-level dropout, completely at random or by baseline severity	500
5	Patient slope variance (SD 0.3 per week), Hybrid, $N = 70$	1,000
S	Hybrid run with independent seeds: G8, G9 and G10, $N \in \{35, 70\}$	1,000 to 5,000

Table 2: The ten G-family analysis specifications. G1 to G3 are the model-based representations of residual exposure. G4 replaces the assumed half-life by a data-driven choice. G6 to G9 are CR2 cluster-robust variants of G1, G2, G4 and G3. G10 is G9 with the half-life chosen on the model without the interaction. G2 and G7 are the lag-adjusted pair, fitted and not discussed. G5 collapses to one row per patient before fitting and has no CR2 variant. The starting analysis E0 is defined in the full report.

Code	Stored	Name	Formula	Target coefficient
G1	E1	Unadjusted	Sx bm + t + Db + bm:Db	bm:Db
G2	E3	Lag-adjusted	Sx bm + t + Db + bm:Db + L	bm:Db
G3	E2	Exposure-weighted	Sx bm + t + Dbc + bm:Dbc	bm:Dbc
G4	E7	AIC-selected t1/2	G3 formula, t1/2 by AIC over {0.25,0.5,1.0,2.0}	bm:Dbc
G5	E9	Paired-difference	diff bm (patient-level OLS)	bm slope
G6	E1cr2	Unadjusted + CR2	G1 formula, CR2 SE	bm:Db
G7	E3cr2	Lag-adjusted + CR2	G2 formula, CR2 SE	bm:Db
G8	E2cr2	Exposure-weighted + CR2	G3 formula, CR2 SE	bm:Dbc
G9	E7cr2	AIC-selected t1/2 + CR2	G4 formula, CR2 SE	bm:Dbc
G10	E7null_cr2	Null-model AIC t1/2 + CR2	G3 formula, t1/2 by AIC on Sx bm + t + Dbc, CR2 SE	bm:Dbc

Table 3: The analysis ladder and its side analysis. Rungs 0 to 4 form the ladder. The optional rows are the model-based counterparts of G8 and G9 and separate a change of regressor from a change of inference. G3 and G8 are oracle benchmarks at the true half-life. The side row, G9, chooses the half-life with the interaction in the model. Sx is the symptom score, bm the biomarker, Db the binary on-drug indicator, Dbc the exposure-weighted predictor and De the design expectancy. Every analysis has a patient random intercept.

Analysis	Rung	Formula	Inference	Assumptions
E0	0	Sx ∼ bm + t + Db + bm:Db (+ De where expectancy varies); random intercept	lme4 with lmerTest Satterthwaite df, no residual correlation	Binary drug indicator; independent residuals given the random intercept; patients never on drug dropped
E0n	isolation	E0 formula	Software and degrees-of-freedom method of the new code	Isolation run: differs from E0 in software and df only
E0nc	isolation	E0 formula with corCAR1 residuals	As E0n	Isolation run: adds residual correlation to E0n
E1nc	isolation	Sx ∼ bm + t + Db + bm:Db; random intercept, no residual correlation	As E0n	Isolation run: removes the expectancy term and patient dropping from E0n
G1	1	Sx ∼ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	nlme model-based standard error	Binary drug indicator; AR(1) residual correlation at the stated working structure
G6	2	Sx ∼ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Binary drug indicator; working covariance may be wrong
G3	optional, 2 to 3	Sx ∼ bm + t + Dbc + bm:Dbc; random intercept, corCAR1 residuals	nlme model-based standard error	Exponential carryover at the true half-life; working covariance correct
G8	3	Sx ∼ bm + t + Dbc + bm:Dbc; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Exponential carryover at the true half-life
G4	optional, 3 to 4	G3 formula with the half-life chosen by AIC over {0.25, 0.5, 1, 2} weeks	nlme model-based standard error, selection ignored	Exponential carryover; true half-life near the candidate grid
G10	4	G8 formula; half-life chosen by ML-AIC over {0.25, 0.5, 1, 2} weeks on the model without bm:Dbc, then refit by REML with it	CR2 cluster-robust standard error, Satterthwaite df, selection ignored	Exponential carryover; true half-life near the candidate grid; selection does not use the interaction term
G9	side	G10 formula with the half-life chosen on the model with bm:Dbc	CR2 cluster-robust standard error, Satterthwaite df, selection ignored	As G10; the selection uses the interaction term

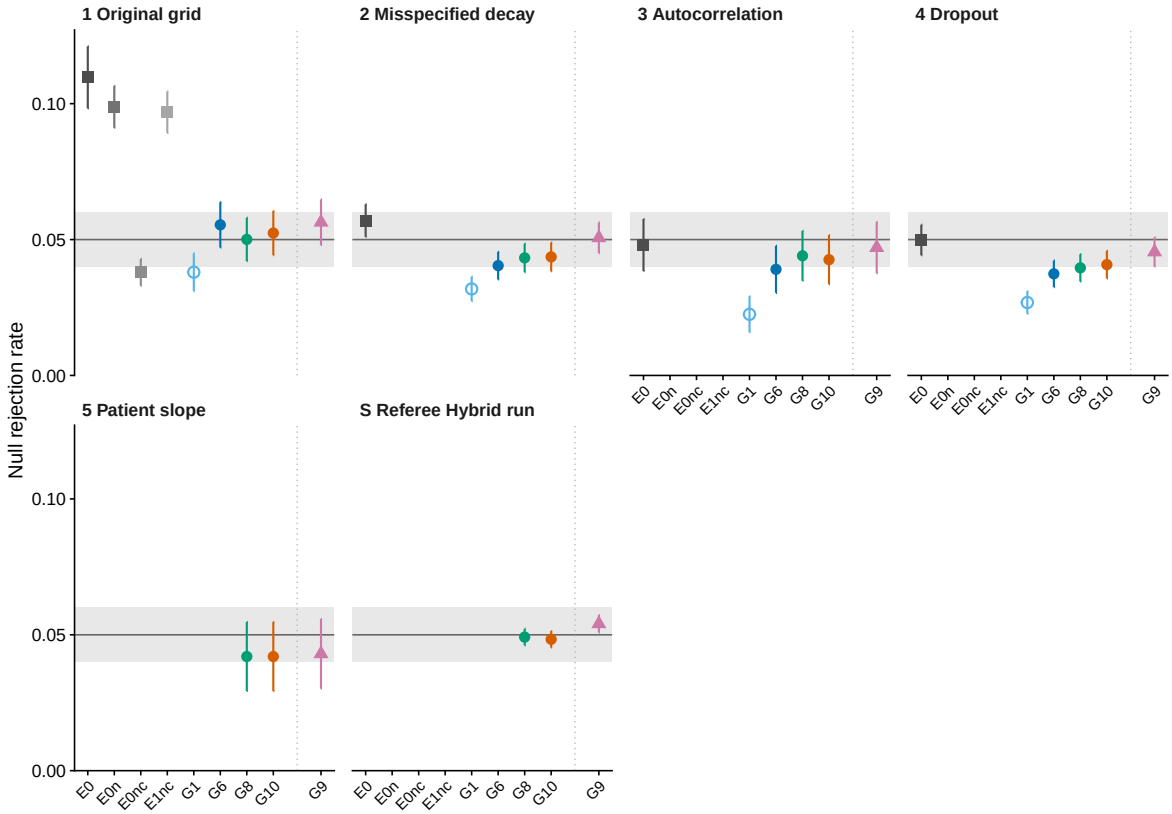


Figure 1: The size ladder. Each panel is one scenario family (Table 1). The vertical axis is the null rejection rate pooled over the null cells of the family, weighted by replicates. The horizontal axis lists the analyses in rung order: E0, G1, G6, G8 and G10 (rungs 0, 1, 2, 3 and 4). The isolation analyses E0n, E0nc and E1nc stand between E0 and G1 in family 1 only (the full report). Right of the dotted line is the side analysis G9. Bars are 2 × the Monte Carlo standard error. The shaded band is 0.04 to 0.06, the two-standard-error band for roughly 2,000 replicates. It is a fixed tolerance around 0.05 and not a test. An analysis that was not fitted in a family has no point.

Table 4: Paired increments along the ladder. Each cell is the later analysis minus the earlier one, with its paired standard error in parentheses, for size (pooled null rejection rate) and power (pooled reference alternative), by scenario family. A dash means the step has no data in that family. The standard errors are conservative because cells within a job share random-number streams. The step from E0 to G1 changes three things at once. The rung 1 isolation steps change one of them at a time. They come from a separate run with 1,000 replicates per cell, so they need not add up to the core step from E0 to G1, which uses job 27. The optional steps separate the change of regressor from the change of inference. The side step compares G9 with G10.

		Scenario family					
Step	Measure	1	2	3	4	5	S
Core steps							
E0 to G1	Size	-0.072 (0.013)	-0.025 (0.010)	-0.025 (0.009)	-0.023 (0.009)	–	–
	Power	-0.128 (0.018)	-0.027 (0.017)	-0.002 (0.017)	-0.017 (0.015)	–	–
G1 to G6	Size	+0.018 (0.008)	+0.009 (0.006)	+0.017 (0.007)	+0.011 (0.007)	–	–
	Power	+0.024 (0.014)	+0.015 (0.012)	+0.044 (0.012)	+0.016 (0.012)	–	–
G6 to G8	Size	-0.006 (0.007)	+0.003 (0.007)	+0.005 (0.006)	+0.002 (0.006)	–	–
	Power	+0.053 (0.013)	+0.013 (0.012)	+0.010 (0.010)	+0.019 (0.009)	–	–
G8 to G10	Size	+0.002 (0.005)	-0.000 (0.004)	-0.002 (0.004)	+0.001 (0.005)	+0.000 (0.003)	-0.001 (0.002)
	Power	-0.011 (0.008)	+0.003 (0.008)	-0.007 (0.008)	-0.013 (0.007)	-0.004 (0.003)	-0.013 (0.006)
Rung 1 isolation steps							
E0 to E0n	Size	+0.000 (0.000)	–	–	–	–	–
	Power	+0.000 (0.000)	–	–	–	–	–
E0n to E0nc	Size	-0.061 (0.008)	–	–	–	–	–
	Power	-0.124 (0.013)	–	–	–	–	–
E0n to E1nc	Size	-0.002 (0.002)	–	–	–	–	–
	Power	-0.005 (0.002)	–	–	–	–	–
E1nc to G1	Size	-0.060 (0.008)	–	–	–	–	–
	Power	-0.129 (0.013)	–	–	–	–	–
Optional model-based steps							
G1 to G3	Size	-0.001 (0.007)	-0.001 (0.006)	+0.003 (0.005)	-0.002 (0.005)	–	–
	Power	+0.063 (0.014)	+0.007 (0.012)	+0.020 (0.010)	+0.015 (0.009)	–	–
G3 to G8	Size	+0.013 (0.007)	+0.013 (0.007)	+0.019 (0.007)	+0.015 (0.007)	–	–
	Power	+0.014 (0.013)	+0.021 (0.013)	+0.034 (0.012)	+0.020 (0.011)	–	–
G3 to G4	Size	+0.005 (0.004)	+0.006 (0.004)	+0.001 (0.004)	+0.005 (0.004)	–	–
	Power	+0.003 (0.008)	+0.017 (0.008)	+0.001 (0.008)	+0.002 (0.006)	–	–
G4 to G9	Size	+0.015 (0.007)	+0.014 (0.007)	+0.021 (0.007)	+0.016 (0.007)	–	–
	Power	+0.013 (0.013)	+0.018 (0.013)	+0.036 (0.012)	+0.016 (0.011)	–	–
Side analysis							
G10 to G9	Size	+0.004 (0.003)	+0.007 (0.003)	+0.005 (0.003)	+0.005 (0.003)	+0.001 (0.001)	+0.006 (0.001)
	Power	+0.012 (0.005)	+0.011 (0.004)	+0.010 (0.004)	+0.011 (0.004)	+0.003 (0.002)	+0.024 (0.005)

G10 lies in the 0.04 to 0.06 band in 6 of 6 families; E0 lies outside it in 1 of 4. Figure 2 gives power and the paired change at each step.

Table 4 gives the paired increments. In the base grid, choosing the half-life on the model without the interaction changes power against the oracle by -0.011 (paired SE 0.008).

Rung 0 to 1: residual correlation

Table 5 gives Type I error at the six null cells of the corrected grid.

Table 6 separates the three differences between E0 and G1.

Rung 2: CR2 inference

Table 7 shows the model-based test and its CR2 correction at four Hybrid null cells.

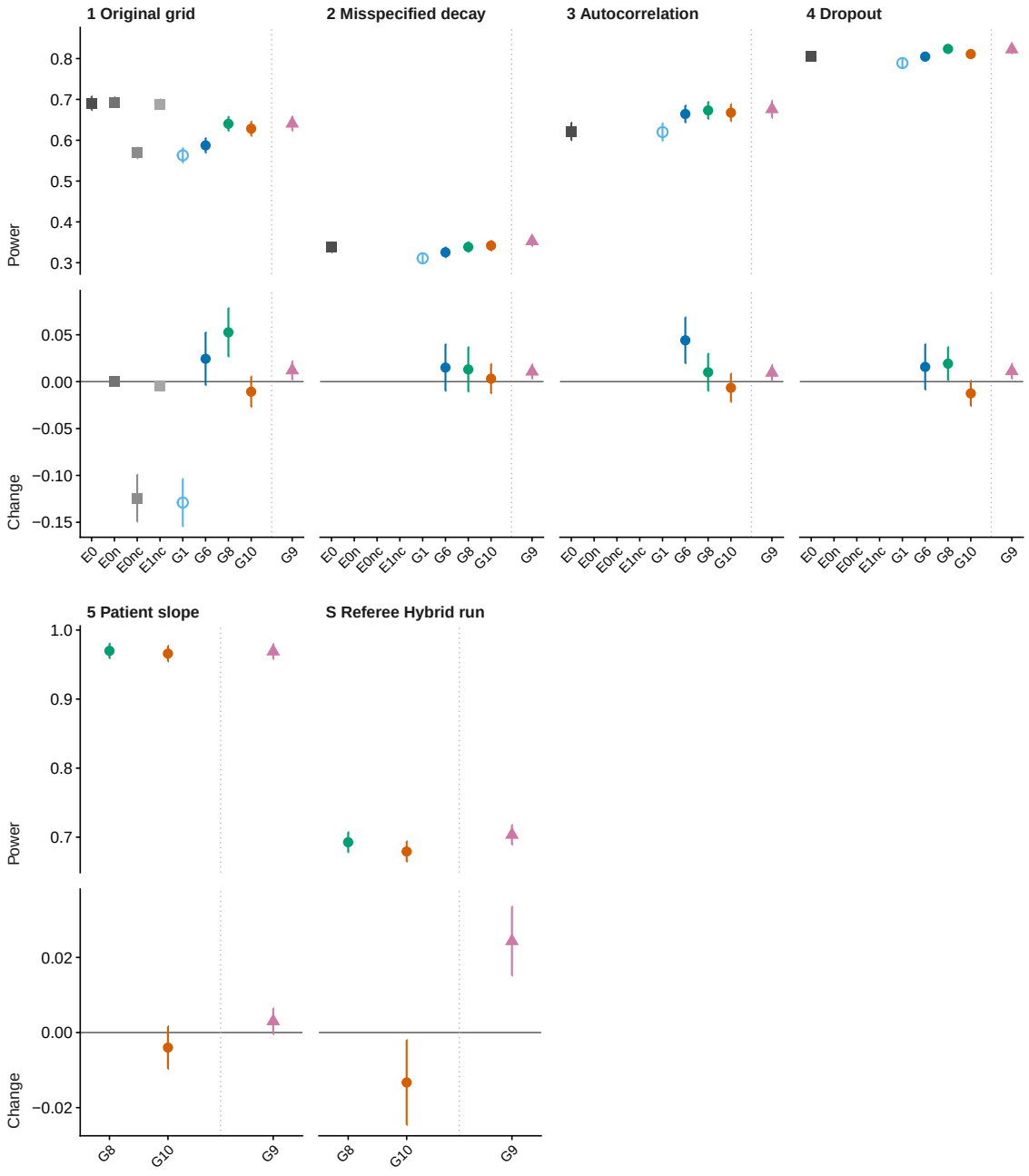


Figure 2: The power ladder. Each panel is one scenario family. The upper part of a panel is the power at the reference alternative, pooled within the family and weighted by replicates. The lower part is the paired change in power from the preceding analysis on the same trials. The preceding analysis is E0 for G1, G1 for G6, G6 for G8, G8 for G10, and G10 for G9. In family 1 the isolation run replaces the step from E0 to G1: E0 precedes E0n, E0n precedes E0nc and E1nc, and E1nc precedes G1. The horizontal axis is in rung order as in Figure 1. Bars are 2 × the Monte Carlo standard error, and the paired standard error for the changes. Paired standard errors are conservative because cells within a job share random-number streams.

Table 5: Type I error at the six null cells of the corrected grid ($c_{bm} = 0$, $t_{1/2} = 1$ week, exponential DGP, $n_{sim} = 500$). Each model-based specification is followed by its CR2 variant (G1 and G6, G3 and G8, G4 and G9). E0 is the starting analysis. Two-MCSE band around 0.05: $[0.031, 0.069]$.

Design	N	G1	G6	G3	G8	G4	G9	E0
CO	35	0.030	0.046	0.034	0.052	0.042	0.058	0.150
CO	70	0.046	0.054	0.046	0.050	0.050	0.056	0.128
Hybrid	35	0.042	0.066	0.030	0.044	0.042	0.060	0.070
Hybrid	70	0.032	0.046	0.034	0.054	0.038	0.058	0.060
OL+BDC	35	0.048	0.074	0.048	0.052	0.044	0.056	0.144
OL+BDC	70	0.030	0.046	0.030	0.048	0.032	0.050	0.106

Table 6: Isolation of the three differences between E0 and G1 (simrun job 42, 1,000 replicates per cell, $t_{1/2} = 1$ week, pooled over $N \in \{35, 70\}$, Monte Carlo standard error in parentheses). E0 is the starting analysis. E0n refits it with nlme and no residual correlation. E0nc adds corCAR1 to E0n. E1nc uses the G1 formula with nlme and no residual correlation, that is, without the expectancy term and without the exclusion rule. G1 is E1nc with corCAR1. Power is at $c_{bm} = 0.45$. The calibration ratio κ is the mean model standard error over the empirical standard deviation, averaged over the two sample sizes. Under CO the expectancy term is absent from E0 because expectancy is constant, so E0n and E1nc are the same model.

Design	Quantity	E0	E0n	E0nc	E1nc	G1
CO	Type I error	0.137 (0.008)	0.137 (0.008)	0.047 (0.005)	0.137 (0.008)	0.046 (0.005)
CO	κ under the null	0.77	0.77	1.00	0.77	1.00
CO	Power	0.821 (0.009)	0.821 (0.009)	0.679 (0.010)	0.821 (0.009)	0.678 (0.010)
Hybrid	Type I error	0.061 (0.005)	0.061 (0.005)	0.034 (0.004)	0.060 (0.005)	0.034 (0.004)
Hybrid	κ under the null	0.95	0.95	1.05	0.96	1.05
Hybrid	Power	0.702 (0.010)	0.702 (0.010)	0.629 (0.011)	0.700 (0.010)	0.631 (0.011)
OL+BDC	Type I error	0.098 (0.007)	0.098 (0.007)	0.034 (0.004)	0.093 (0.007)	0.031 (0.004)
OL+BDC	κ under the null	0.83	0.83	1.07	0.85	1.09
OL+BDC	Power	0.557 (0.011)	0.557 (0.011)	0.399 (0.011)	0.545 (0.011)	0.369 (0.011)

Table 7: Conservatism of the model-based interaction test and its cluster-robust correction at four Hybrid null cells ($c_{bm} = 0$, exponential DGP, corrected simulation code, 2,000 replicates each). κ is the calibration ratio of the full report. $\kappa > 1$ is a conservative test. The CR2 columns are G6, G8 and G9 for G1, G3 and G4 respectively. Two-MCSE band around 0.05: $[0.040, 0.060]$.

$t_{1/2}$	N	Specification	κ model	Type I model	κ CR2	Type I CR2
0.5	35	G1	1.05	0.041	0.97	0.049
0.5	35	G3	1.05	0.035	0.97	0.050
0.5	35	G4	1.00	0.043	0.93	0.056
0.5	70	G1	1.05	0.042	0.98	0.050
0.5	70	G3	1.06	0.037	0.99	0.050
0.5	70	G4	1.05	0.040	0.98	0.052
1.0	35	G1	1.05	0.040	0.97	0.047
1.0	35	G3	1.06	0.033	0.98	0.052
1.0	35	G4	0.99	0.046	0.91	0.059
1.0	70	G1	1.05	0.041	0.98	0.049
1.0	70	G3	1.07	0.035	0.99	0.046
1.0	70	G4	1.03	0.042	0.96	0.055

Figures 3 and 4 give power for the nine starting specifications on the corrected grid.

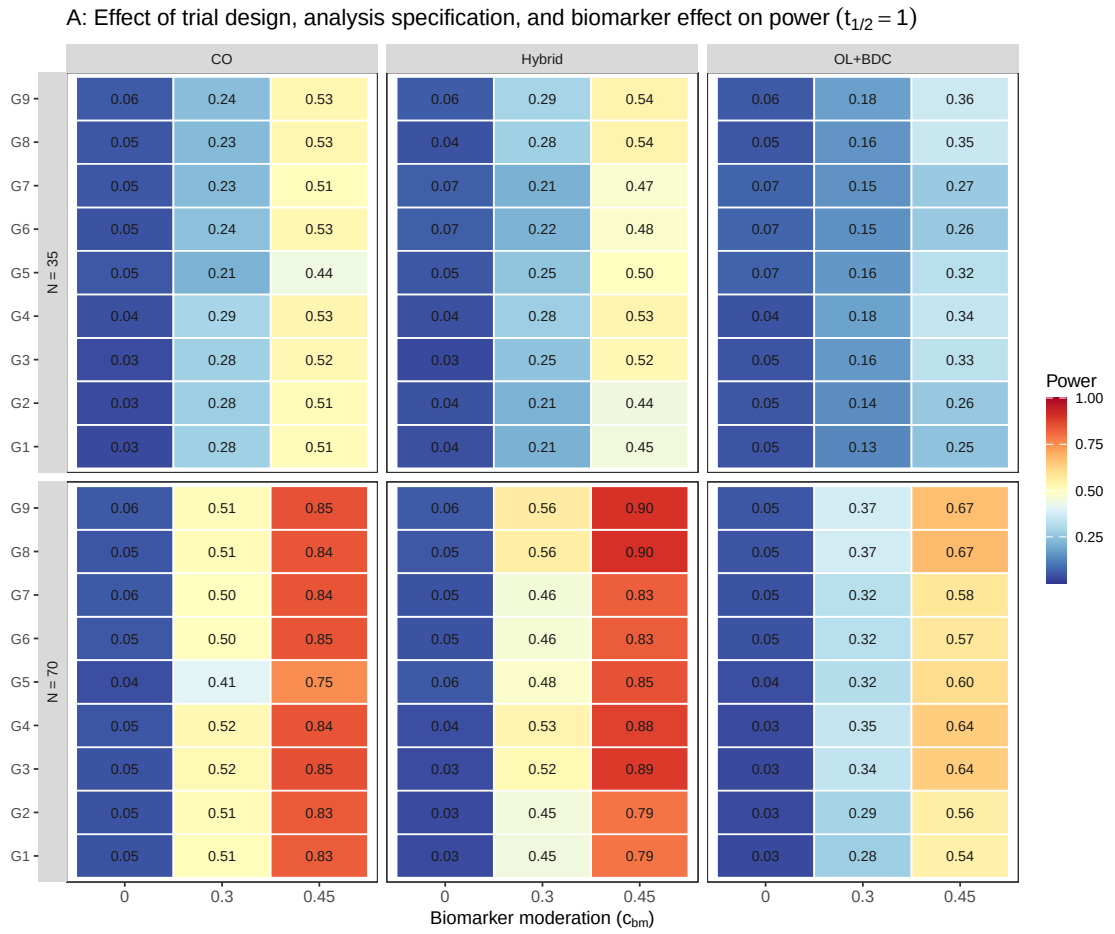


Figure 3: Power by trial design (columns), N (row blocks) and analysis specification (rows within block) across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, corrected simulation code ($n_{sim} = 500$). The $c_{bm} = 0$ column is Type I error.

Rung 3: exposure weighting

Table 8 gives the reference cell and Figure 5 power against the half-life.

Rung 4: choosing the half-life

Table 9 compares G1, G3 and G4 (model-based).

Tables 10 and 11 give the size cost of selecting with the interaction in the model and the selected half-lives.

Tables 12 and 13 compare G8, G10 and G9 on the same trials in the independent Hybrid run.

G10 differs from G8 under the null by -0.0012 to -0.0002; G9 exceeds G8 by +0.0024 to +0.0088.

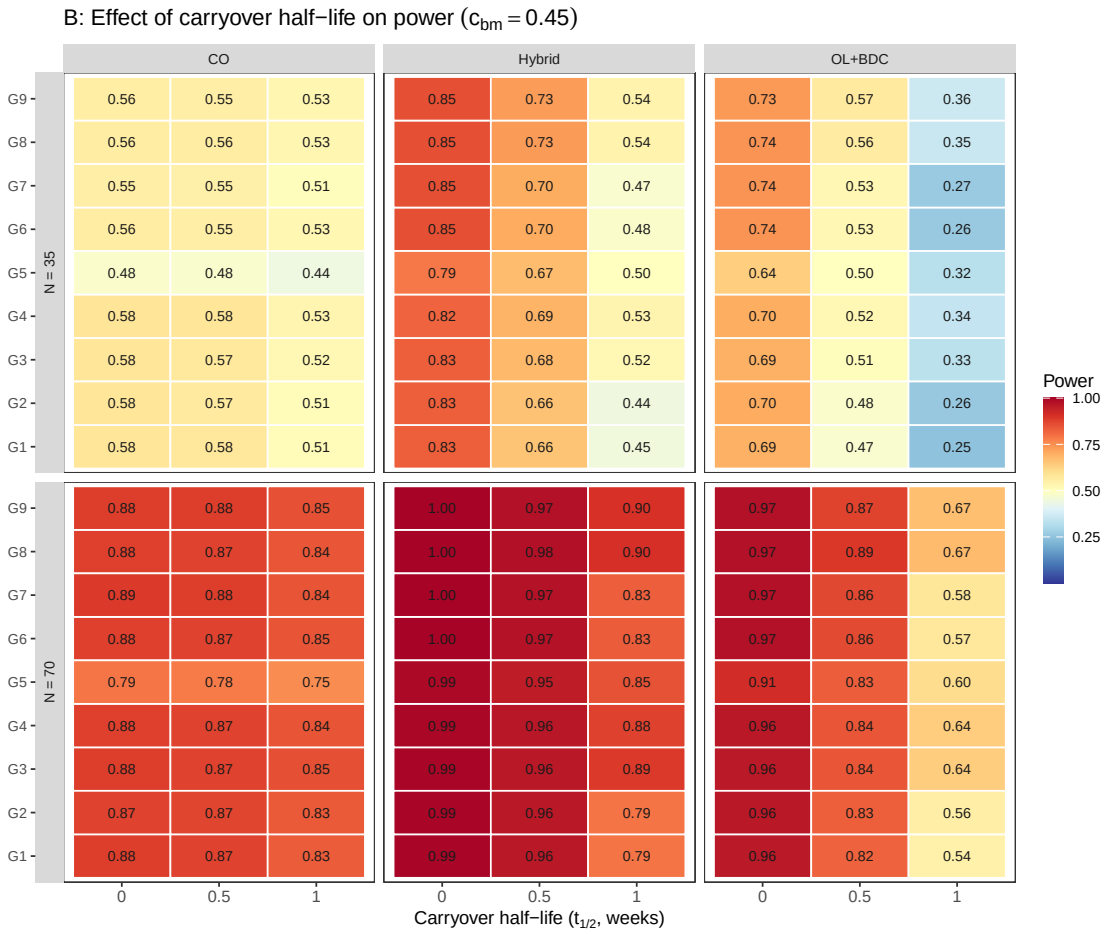


Figure 4: Power across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, corrected simulation code ($n_{sim} = 500$).

Table 8: Reference cell on the corrected 30-cell grid (job 27: Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates, every analysis fitted to the same trials). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$ with its Monte Carlo standard error. Type I is the rejection rate in the null cell of the same design and N. That cell exists only at $t_{1/2} = 1$ week. Empirical SD is the standard deviation of the point estimates and Mean SE the mean model standard error. κ is their ratio. Bias is the mean estimate minus $\theta = -c_{bm} \cdot 8 = -3.6$ and is shown only for the exposure-weighted analyses G3, G8, G4 and G9. G1 and G6 estimate the coefficient of the binary indicator, which is a different quantity, and E0 is shown for power and size only. G1, G3 and G4 use model-based inference. G6, G8 and G9 use CR2. Non-conv is the fraction of fits that did not converge or gave no p-value.

$t_{1/2}$	Analysis	Power	Type I	Empirical SD	Mean SE	κ	Bias	Non-conv
0.0	G1	0.992 (0.004)	–	0.73	0.88	1.21	–	0.000
	G6	0.996 (0.003)	–	0.73	0.71	0.98	–	0.004
	G3	0.992 (0.004)	–	0.73	0.88	1.21	-0.03	0.000
	G8	0.996 (0.003)	–	0.73	0.71	0.98	-0.03	0.004
	G4	0.990 (0.004)	–	0.75	0.91	1.21	-0.16	0.000
	G9	0.996 (0.003)	–	0.75	0.74	0.98	-0.16	0.000
	E0	0.986 (0.005)	–	–	–	–	–	0.000
0.5	G1	0.960 (0.009)	–	0.76	0.88	1.16	–	0.000
	G6	0.972 (0.007)	–	0.76	0.75	0.99	–	0.002
	G3	0.960 (0.009)	–	0.86	1.02	1.18	-0.02	0.000
	G8	0.976 (0.007)	–	0.86	0.85	0.99	-0.02	0.006
	G4	0.962 (0.009)	–	0.87	0.97	1.11	+0.14	0.006
	G9	0.974 (0.007)	–	0.87	0.82	0.94	+0.14	0.006
	E0	0.950 (0.010)	–	–	–	–	–	0.000
1.0	G1	0.790 (0.018)	0.032 (0.008)	0.79	0.89	1.12	–	0.000
	G6	0.831 (0.017)	0.046 (0.009)	0.79	0.78	0.99	–	0.004
	G3	0.886 (0.014)	0.034 (0.008)	1.05	1.22	1.15	-0.02	0.000
	G8	0.904 (0.013)	0.054 (0.010)	1.05	1.05	1.00	-0.02	0.000
	G4	0.876 (0.015)	0.038 (0.009)	1.29	1.21	0.94	-0.06	0.004
	G9	0.900 (0.013)	0.058 (0.010)	1.29	1.05	0.82	-0.06	0.004
	E0	0.846 (0.016)	0.060 (0.011)	–	–	–	–	0.000

Table 9: Power of the binary indicator (G1), the exposure-weighted predictor at the true half-life (G3) and the exposure-weighted predictor at the AIC-selected half-life (G4), model-based inference, on the 30-cell corrected grid (exponential DGP, $n_{sim} = 500$, same simulated trial for all three each replicate). Two-MCSE half-width is about 0.04 at $p = 0.5$.

Design	N	c_{bm}	$t_{1/2}$	G1	G3	G4
CO	35	0.45	0.0	0.582	0.582	0.582
CO	35	0.45	0.5	0.576	0.572	0.580
CO	35	0.30	1.0	0.280	0.280	0.294
CO	35	0.45	1.0	0.514	0.524	0.532
CO	70	0.45	0.0	0.880	0.880	0.878
CO	70	0.45	0.5	0.874	0.874	0.874
CO	70	0.30	1.0	0.508	0.518	0.522
CO	70	0.45	1.0	0.832	0.852	0.844
Hybrid	35	0.45	0.0	0.826	0.826	0.816
Hybrid	35	0.45	0.5	0.656	0.678	0.688
Hybrid	35	0.30	1.0	0.212	0.250	0.276
Hybrid	35	0.45	1.0	0.450	0.524	0.534
Hybrid	70	0.45	0.0	0.992	0.992	0.990
Hybrid	70	0.45	0.5	0.960	0.960	0.962
Hybrid	70	0.30	1.0	0.448	0.518	0.534
Hybrid	70	0.45	1.0	0.790	0.886	0.876
OL+BDC	35	0.45	0.0	0.694	0.694	0.700
OL+BDC	35	0.45	0.5	0.474	0.506	0.523
OL+BDC	35	0.30	1.0	0.134	0.164	0.182
OL+BDC	35	0.45	1.0	0.248	0.330	0.344
OL+BDC	70	0.45	0.0	0.958	0.958	0.960
OL+BDC	70	0.45	0.5	0.820	0.842	0.842
OL+BDC	70	0.30	1.0	0.276	0.338	0.346
OL+BDC	70	0.45	1.0	0.544	0.638	0.639

Table 10: Null rejection of G6, G8 and G9 pooled over the four 2,000-replicate Hybrid null cells of Table 7; the paired difference between G9 and G8 on the same trials (the discordant counts are 77 replicates rejected by G9 only and 30 by G8 only); and G9 conditional on the half-life it selected. From '34-referee-reanalysis.R'.

Quantity	Method	n	Rate	MCSE
Pooled null rejection	G6	7986	0.0486	0.0024
Pooled null rejection	G8	7988	0.0493	0.0024
Pooled null rejection	G9	7987	0.0555	0.0026
Paired difference, G9 minus G8	G9-G8	7980	0.0059	0.0013
G9 given selected $\hat{t}^a_{1/2} = 0.25$	G9	3116	0.0507	0.0039
G9 given selected $\hat{t}^a_{1/2} = 0.5$	G9	2232	0.0587	0.0050
G9 given selected $\hat{t}^a_{1/2} = 1$	G9	1613	0.0583	0.0058
G9 given selected $\hat{t}^a_{1/2} = 2$	G9	1026	0.0585	0.0073

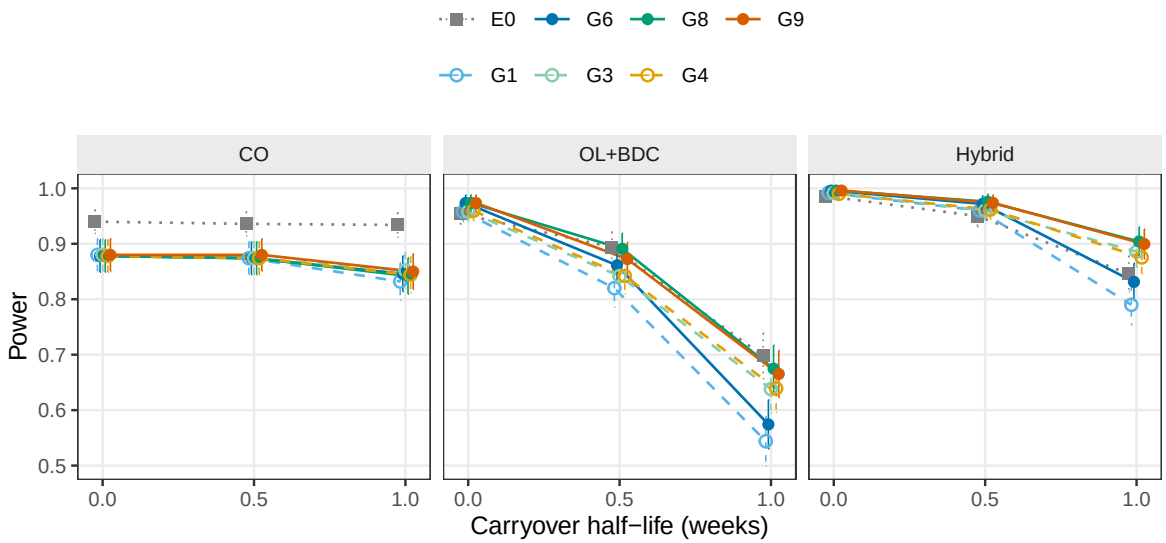


Figure 5: Power against carryover half-life by analysis and design (job 27: $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates). The analyses are G1 (binary indicator, model-based), G6 (binary indicator, CR2), G3 (exposure-weighted at the true half-life, model-based), G8 (the same with CR2), G4 (half-life chosen by AIC, model-based), G9 (the same with CR2) and E0, the starting analysis, in gray. Points at each half-life are offset horizontally for legibility. Bars are 2 × the Monte Carlo standard error.

Table 11: Share of replicates in which each candidate half-life (weeks) was selected by AIC, Hybrid design, pooled over every stored cell at the given true half-life and N (null and alternative cells together). From '34-referee-reanalysis.R'.

			Selected half-life			
True $t_{1/2}$	N	n	0.25	0.5	1	2
0.0	35	500	0.97	0.02	0.00	0.00
0.0	70	500	0.99	0.01	0.00	0.00
0.2	35	1000	0.95	0.05	0.00	0.00
0.2	70	992	0.98	0.02	0.00	0.00
0.5	35	34957	0.56	0.29	0.11	0.04
0.5	70	47337	0.59	0.33	0.06	0.01
1.0	35	3500	0.20	0.24	0.29	0.28
1.0	70	3493	0.12	0.26	0.40	0.21
2.0	35	1000	0.16	0.07	0.14	0.63
2.0	70	1992	0.06	0.06	0.15	0.73
4.0	35	1000	0.26	0.06	0.07	0.60
4.0	70	991	0.18	0.03	0.07	0.72

Table 12: Referee-variant runs: null rejection rate ($c_{bm} = 0$) of G8 (the oracle at the true half-life), G10 and G9 in four Hybrid null cells, 5,000 replicates each (MCSE about 0.003), independent seeds per cell, with the paired per-replicate difference against G8 (paired SE 0.0015 to 0.0021) and the calibration ratio.

Cell	Method	Rejection	Paired diff. vs G8	κ
$t_{1/2} = 0.5$, $N = 35$	G8	0.0482	–	0.97
	G10	0.0470	-0.0012	0.97
	G9	0.0522	+0.0040	0.93
$t_{1/2} = 0.5$, $N = 70$	G8	0.0514	–	0.99
	G10	0.0510	-0.0004	0.98
	G9	0.0538	+0.0024	0.97
$t_{1/2} = 1.0$, $N = 35$	G8	0.0508	–	0.96
	G10	0.0506	-0.0002	0.94
	G9	0.0596	+0.0088	0.89
$t_{1/2} = 1.0$, $N = 70$	G8	0.0461	–	1.00
	G10	0.0447	-0.0012	0.99
	G9	0.0505	+0.0044	0.95

Table 13: Referee-variant runs: power at eight Hybrid alternative cells, 1,000 replicates each (MCSE at most 0.016), independent seeds per cell, for the same analyses as the null table. The true half-life is 0.5 or 2 weeks.

$t_{1/2}$	N	c_{bm}	G8	G10	G9
0.5	35	0.25	0.291	0.277	0.293
0.5	70	0.25	0.538	0.526	0.538
0.5	35	0.45	0.743	0.737	0.753
0.5	70	0.45	0.966	0.960	0.964
2.0	35	0.25	0.124	0.108	0.139
2.0	70	0.25	0.248	0.239	0.271
2.0	35	0.45	0.384	0.362	0.410
2.0	70	0.45	0.679	0.661	0.690

Table 14: Type I error ($c_{bm} = 0$) at four null scenarios on Hybrid ($N = 70$, $t_{1/2} = 0.5$ week unless stated; OL+BDC row is that design), $n_{sim} = 500$, same trial for every specification each replicate. G10 was fitted to the same trials of the two misspecification cells only (– elsewhere). Two-MCSE band around 0.05: $[0.031, 0.069]$.

Cell	G1	G3	G4	G6	G8	G10	G9	E0
Hybrid	0.034	0.032	0.034	0.052	0.046	–	0.048	0.060
OL+BDC	0.030	0.036	0.036	0.051	0.048	–	0.060	0.100
$t_{1/2} = 2$ wk	0.034	0.038	0.042	0.052	0.054	0.056	0.062	0.066
Weibull $k = 2$ data	0.032	0.032	0.032	0.052	0.046	0.052	0.052	0.060

Table 15: Power at five alternative scenarios (same specifications and n_{sim} as the null table). G6 and G8 are the equal-size comparators; E0 is liberal and its power is not comparable.

Cell	G1	G3	G4	G6	G8	G10	G9	E0
Hybrid, $c_{bm} = 0.45$	0.934	0.950	0.950	0.962	0.972	–	0.968	0.932
Hybrid, $c_{bm} = 0.25$	0.510	0.542	0.544	0.564	0.574	–	0.584	0.546
OL+BDC, $c_{bm} = 0.45$	0.842	0.842	0.825	0.865	0.871	–	0.863	0.866
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	0.180	0.242	0.265	0.196	0.257	0.253	0.281	0.284
Weibull $k = 2$ data, $c_{bm} = 0.25$	0.630	0.610	0.631	0.653	0.655	0.661	0.661	0.612

Stress scenarios

Tables 14 and 15 give the misspecification cells of the scenario job.

Tables 16 and 17 cross the scenarios with two sample sizes. The G10 Type I error is 0.032 to 0.052.

Table 18 imposes monotone dropout.

The G10 Type I error under dropout is 0.026 to 0.054. Table 19 is the heavy-missingness case.

Table 20 adds a patient-specific linear trend.

Calibration and coverage

Figure 6 gives the calibration ratio and Table 21 the coverage of G8 and G9.

Which analysis to use

Table 22 scores the analyses over the families.

Recommendation

For a Hybrid trial with a predictive-only biomarker at $N \leq 70$ we recommend G10: the exposure-weighted predictor with the half-life chosen by AIC on the model without the interaction, refitted by REML with the interaction and tested with CR2. Against the oracle at

Table 16: Scenario grids: Type I error ($c_{bm} = 0$) on Hybrid at $N \in \{35, 70\}$ under a Weibull decay analyzed as exponential, a true half-life below, at the boundary of and above G4's candidate grid, and two autocorrelations ($t_{1/2} = 0.5$ week and $\rho = 0.7$ unless stated; $n_{sim} = 500$; band $[0.031, 0.069]$).

Scenario	N	G1	G3	G4	G6	G8	G10	G9	E0
Weibull $k = 0.5$ data	35	0.038	0.040	0.048	0.040	0.042	0.044	0.052	0.054
Weibull $k = 0.5$ data	70	0.026	0.028	0.028	0.034	0.040	0.044	0.048	0.060
Weibull $k = 2$ data	35	0.038	0.032	0.040	0.040	0.042	0.038	0.038	0.054
Weibull $k = 2$ data	70	0.026	0.028	0.030	0.036	0.042	0.036	0.038	0.062
$t_{1/2} = 0.2$ wk	35	0.038	0.040	0.040	0.040	0.038	0.038	0.038	0.054
$t_{1/2} = 0.2$ wk	70	0.026	0.026	0.028	0.036	0.032	0.034	0.034	0.060
$t_{1/2} = 2$ wk	35	0.036	0.022	0.034	0.042	0.050	0.044	0.056	0.048
$t_{1/2} = 2$ wk	70	0.026	0.030	0.034	0.034	0.046	0.050	0.058	0.056
$t_{1/2} = 4$ wk	35	0.036	0.022	0.044	0.042	0.044	0.042	0.064	0.050
$t_{1/2} = 4$ wk	70	0.026	0.026	0.038	0.036	0.042	0.044	0.067	0.060
$\rho = 0.5$	35	0.032	0.036	0.040	0.044	0.046	0.052	0.060	0.056
$\rho = 0.5$	70	0.024	0.030	0.028	0.030	0.044	0.040	0.046	0.062
$\rho = 0.9$	35	0.016	0.018	0.016	0.038	0.036	0.032	0.034	0.040
$\rho = 0.9$	70	0.018	0.018	0.022	0.044	0.050	0.046	0.048	0.034

Table 17: Scenario grids: power at one alternative per scenario (same cells and n_{sim} as the null table; $c_{bm} = 0.30$ for the autocorrelation rows because 0.45 is infeasible at $\rho = 0.9$).

Scenario	N	G1	G3	G4	G6	G8	G10	G9	E0
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	35	0.538	0.552	0.572	0.548	0.586	0.566	0.588	0.586
Weibull $k = 0.5$ data, $c_{bm} = 0.45$	70	0.878	0.894	0.889	0.900	0.915	0.907	0.913	0.876
Weibull $k = 2$ data, $c_{bm} = 0.45$	35	0.830	0.814	0.828	0.850	0.848	0.850	0.850	0.794
Weibull $k = 2$ data, $c_{bm} = 0.45$	70	0.994	0.990	0.994	0.996	0.994	0.994	0.994	0.986
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	35	0.348	0.344	0.342	0.340	0.342	0.338	0.338	0.336
$t_{1/2} = 0.2$ wk, $c_{bm} = 0.25$	70	0.632	0.626	0.628	0.659	0.659	0.655	0.657	0.614
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	35	0.108	0.124	0.150	0.112	0.146	0.150	0.168	0.160
$t_{1/2} = 2$ wk, $c_{bm} = 0.25$	70	0.176	0.220	0.235	0.185	0.233	0.234	0.254	0.246
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	35	0.058	0.066	0.080	0.064	0.080	0.090	0.102	0.098
$t_{1/2} = 4$ wk, $c_{bm} = 0.25$	70	0.084	0.104	0.137	0.092	0.134	0.113	0.139	0.140
$\rho = 0.5$, $c_{bm} = 0.30$	35	0.280	0.300	0.310	0.284	0.288	0.286	0.306	0.358
$\rho = 0.5$, $c_{bm} = 0.30$	70	0.542	0.546	0.552	0.595	0.606	0.585	0.596	0.610
$\rho = 0.9$, $c_{bm} = 0.30$	35	0.684	0.734	0.717	0.785	0.808	0.804	0.808	0.608
$\rho = 0.9$, $c_{bm} = 0.30$	70	0.974	0.980	0.980	0.994	0.990	0.994	0.994	0.910

Table 18: Type I error and power under monotone dropout (Hybrid, $t_{1/2} = 0.5$ week; the rate is the fraction of patients with an incomplete series; $n_{sim} = 500$; band $[0.031, 0.069]$).

	Type I error				Power, $c_{bm} = 0.45$			
Dropout (rate, N)	G1	G8	G10	G9	G1	G8	G10	G9
MAR 0.1, $N = 35$	0.028	0.042	0.044	0.050	0.672	0.719	0.697	0.717
MAR 0.2, $N = 35$	0.034	0.046	0.046	0.052	0.638	0.687	0.669	0.687
MAR 0.3, $N = 35$	0.030	0.044	0.048	0.054	0.596	0.630	0.593	0.617
MCAR 0.1, $N = 35$	0.030	0.048	0.054	0.062	0.674	0.716	0.700	0.718
MCAR 0.2, $N = 35$	0.024	0.040	0.042	0.054	0.644	0.686	0.663	0.681
MCAR 0.3, $N = 35$	0.024	0.036	0.036	0.042	0.608	0.646	0.623	0.649
MAR 0.1, $N = 70$	0.026	0.040	0.038	0.040	0.956	0.980	0.974	0.976
MAR 0.2, $N = 70$	0.022	0.038	0.040	0.042	0.936	0.972	0.974	0.976
MAR 0.3, $N = 70$	0.038	0.040	0.038	0.042	0.910	0.940	0.946	0.948
MCAR 0.1, $N = 70$	0.022	0.036	0.038	0.040	0.958	0.978	0.978	0.978
MCAR 0.2, $N = 70$	0.020	0.032	0.026	0.026	0.950	0.976	0.970	0.970
MCAR 0.3, $N = 70$	0.024	0.032	0.038	0.040	0.926	0.956	0.946	0.952

Table 19: Heavy-missingness stress case (per-occasion MCAR 0.3, $N = 35$, 500 replicates): dispersion of the interaction estimates on the calibrated scale and the calibration ratio on the conventional scale (κ, mean SE over SD) and a robust one (median SE over 1.4826 times the median absolute deviation). From '34-referee-reanalysis.R'.

Cell	Method	SD of estimates	MAD-based SD	Max $\lvert\hat{\beta}\rvert$	κ	Robust κ	Rejection
$c_{bm} = 0.45$	G6	6.1	3.1	89	0.58	0.94	0.075
$c_{bm} = 0.45$	G8	7.5	3.9	110	0.57	0.91	0.085
$c_{bm} = 0.45$	G9	12.4	5.2	137	0.46	0.78	0.083
null	G6	6.1	3.3	82	0.61	0.90	0.020
null	G8	7.6	4.2	102	0.60	0.88	0.024
null	G9	12.4	5.1	131	0.49	0.86	0.022

Table 20: Referee-variant runs: rejection rate (MCSE), mean interaction estimate and calibration ratio under a patient-specific linear trend in the natural-history component (SD 0.3 outcome units per week), Hybrid, $N = 70$, $t_{1/2} = 0.5$ week, 1,000 replicates. G10 was fitted to the same trials by the backfill run. The true value at the alternative is −3.6.

Cell	Method	Rejection	Mean estimate	κ
null ($c_{bm} = 0$)	G8	0.042 (0.006)	-0.02	1.01
	G10	0.042 (0.006)	-0.01	1.00
	G9	0.043 (0.006)	-0.01	0.99
$c_{bm} = 0.45$	G8	0.970 (0.005)	-3.63	0.98
	G10	0.966 (0.006)	-3.39	0.94
	G9	0.969 (0.005)	-3.45	0.92

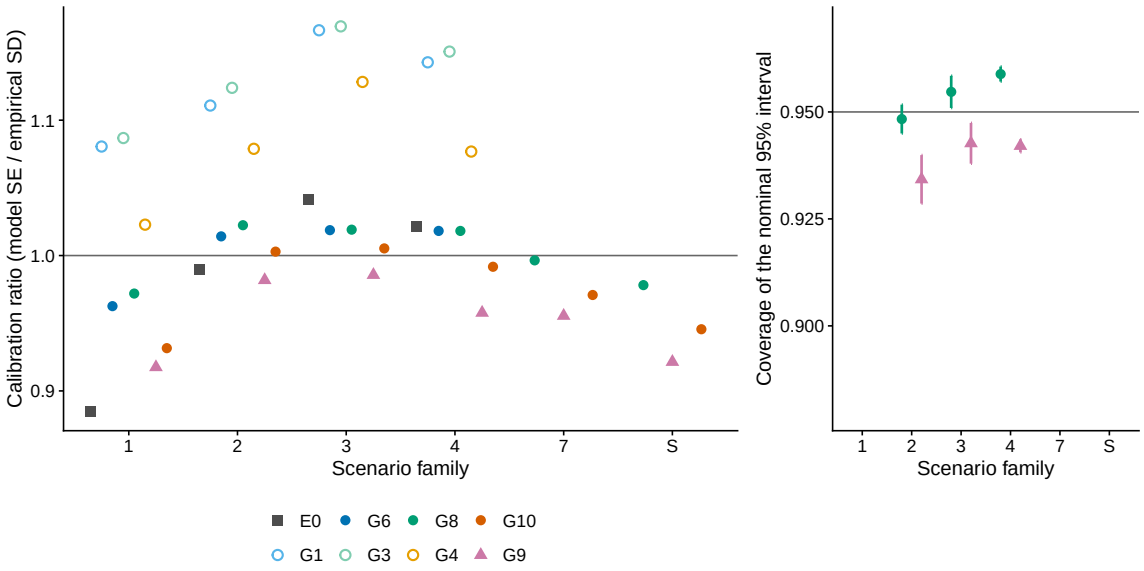


Figure 6: Calibration and coverage. Left: the calibration ratio, the mean model standard error divided by the empirical standard deviation of the estimates, averaged without weights over the cells of each scenario family, by analysis. A value of one is calibrated and a value above one is conservative. Right: the coverage of the nominal 95 percent interval of G8 and G9 in families 2 to 4, as the mean of the cell lower bounds and upper bounds. The bars span the two bounds. The coverage bounds are not Monte Carlo intervals. Coverage was not computed for G10, and families 1, 5 and S have no coverage output.

Table 21: Coverage of the nominal 95 percent CR2 interval for G8 and G9 over the alternative cells of the supplementary study, on the calibrated scale ($\theta = -c_{bm}\sigma_{BR}$). The Satterthwaite degrees of freedom are not stored per replicate and were recovered from the stored t and p where $|t| \geq 0.5$; the remaining replicates are classified as covering, not covering, or ambiguous, which gives the lower and upper bounds. From '34-referee-reanalysis.R'.

Method	N	Cells	Mean coverage (lower, upper)	Minimum (lower)
G8	35	21	0.950, 0.959	0.938
G8	70	21	0.948, 0.953	0.915
G9	35	21	0.938, 0.947	0.890
G9	70	21	0.929, 0.936	0.873

Table 22: Scoreboard of the analyses over the scenario families. Families in band counts the families whose pooled null rejection rate lies in 0.04 to 0.06, out of the families in which the analysis was fitted. Smallest and largest size are cell-level null rejection rates. Power is the mean over families of the pooled power as a percent of G10 in the same family. The worst power loss is against G10 in points of power, by family and by cell; a negative value is a gain. Coverage is the mean of the nominal 95 percent interval over families with coverage output (G8 and G9 only). A dash means not available. G10 is the reference, so its losses are zero.

Analysis	Families in band	Smallest size	Largest size (cell)	Power, percent of G10	Worst family loss, points (family)	Worst cell loss, points (cell)	Coverage
E0	3 of 4	0.034	0.150 (CO, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.3	4.6 (3)	19.7 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G1	0 of 4	0.016	0.048 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	92.7	6.5 (1)	12.1 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G6	2 of 4	0.030	0.074 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	96.9	4.1 (1)	9.3 (OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 1$)	–
G3	0 of 4	0.010	0.048 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	96.9	2.8 (3)	7.1 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G8	5 of 6	0.032	0.054 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.9	0.3 (2)	2.0 (OL+BDC, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 1$)	0.954
G4	1 of 4	0.016	0.050 (CO, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	98.3	2.8 (3)	8.8 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
G10	6 of 6	0.026	0.056 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.0	0	0.0 ($t_{1/2} = 0.2$ wk, $N = 35$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$)	–
G9	6 of 6	0.026	0.067 ($t_{1/2} = 4$ wk, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 4$)	101.9	-0.3 (5)	0.0 ($t_{1/2} = 0.2$ wk, $N = 35$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$)	0.940

the true half-life it changes size by -0.002 to +0.002 and power by -0.013 to +0.003 across the families. Choosing the half-life with the interaction in the model (G9) adds +0.001 to +0.007 of size and +0.003 to +0.024 of power. The recommendation is conditional on the recommended DGP and the parameters simulated.

References

1. pmsimstats team. Prognostic biomarkers and the biomarker-treatment interaction test in aggregated N-of-1 trials. 2026.